

AMENDMENT NO. 1 SEPTEMBER 1977

TO

IS : 2825-1969 CODE FOR UNFIRED PRESSURE VESSELS

Alterations

[Page 18, clause 3.4.2 (c)] — Substitute 'Torispherical' for 'Dished and flanged'.

[Page 19, caption of Fig. 3.3 (C)] — Substitute 'Torispherical Ends' for 'Dished and Flanged Ends'.

(Page 21, clause 3.4.5, Note 2, line 3) — Substitute ' $\frac{d}{\sqrt{t.D_0}}$ ' for ' $d \sqrt{t.D_0}$ ' at both the places.

(Page 24, clause 3.6.1.1, definition of symbol D) — Substitute the following for the existing definition:

' D = diameter or short span measured as in Table 3.7, in mm; '.

[Page 33, clause 3.8.5.2 (b) (3), definition of symbol H_2 , line 6] — Substitute ' $\sqrt{d(t-2c)}$ ' for ' $\sqrt{d(t-2C)}$ '.

(Page 43, clause 4.4, definition of symbol A) — Substitute 'mm' for 'mm²'.

(Page 43, clause 4.4, definition of symbol C_F) — Substitute the following for the existing matter:

' C_F = bolt pitch correction factor

$$= \sqrt{\frac{\text{bolt spacing}}{2(\text{bolt diameter}) + t}}$$

The value of C_F shall be, however, not less than 1.'

(Page 44, clause 4.4) — After the definition of symbol F_L and before the sign of equality '=', insert the notation 'f'.

(Page 44, clause 4.4, definition of symbol G , line 7) — Substitute ' $b_0 > 6.3 \text{ mm}$ ' for ' $b > 6.3 \text{ mm}$ '.

[Page 46, Fig. 4.1 (B)] — Substitute ' h_t ' for ' h_r '.

[Page 46, Fig. 4.1 (E), (F), (G)]

a) Add the following caption above the figures:

'INTEGRAL-TYPE FLANGES'

b) Substitute ' h_t ' for ' h_r ' and ' H_T ' for ' H_r '.

(Page 47, Fig. 4.2) — Substitute ' $b_o \leq 6.3 \text{ mm}$ ' for ' $b_o = 6.3 \text{ mm}$ '.

(Pages 54 and 55, Table 4.2, col I and II) — Substitute ' W ' for ' w ' and ' T ' for ' $25 T$ '.

(Page 67, clause 6.4.2.3, last sentence) — Substitute the following for the existing sentence:

'When plates are flame-cut, the edges should be examined, as far as possible, immediately after this operation.'

(Page 72, clause 6.6.6, second sentence) — Substitute the following for the existing sentence:

'Unless otherwise specified the out of straightness of the shell shall not exceed 0.3 percent of the total cylindrical length in any 5 m length.'

(Page 75, clause 6.12.2.5, line 8) — Substitute 'physical properties' for 'yield point'.

(Page 78, clause 6.12.4.2, line 5) — Substitute 'weld' for 'shell'.

[Page 103, clause 8.7.1.1 (c), line 4] — Substitute '19 mm' for '20 mm'.

(Page 167, Table C.5, last row) — Substitute ' $\frac{b_1}{2}$ ' for ' $\frac{b}{2}$ '.

(Page 167, clause C-4.3.2.4, definition of f_n) — Substitute the following for the existing matter:

' $f_n = pD_o C/2t \times 100$ in which C is the shape factor from Fig. 3.7.'

[Page 168, equations (C.19) and (C.20)] — Substitute ' I ' for ' l ' at both the places.

(Page 181, clause F-4.1, definition of symbol I_s , line 2) — Substitute ' cm^4 ' for ' mm^4 '.

(Pages 115, 116 and 117, Table A.1) — Substitute the following for the existing Table:

TABLE A.1 ALLOWABLE STRESS VALUES FOR CARBON AND LOW ALLOY STEEL IN TENSION

MATERIAL SPECIFICATION	GRADE OR DESIGNATION	CATEGORY OF CHEMICAL COMPOSITION (see TABLE B.1)	MECHANICAL PROPERTIES			ALLOWABLE STRESS VALUES IN kgf/mm ² AT DESIGN TEMPERATURE, °C																
			Tensile Strength Min kgf/mm ² R ₂₀	Yield Stress Min kgf/mm ² E ₂₀	Percentage Elongation Min on Gauge Length = 5.65√S ₀	Up to 50	Up to 100	Up to 150	Up to 200	Up to 250	Up to 300	Up to 350	Up to 375	Up to 400	Up to 425	Up to 450	Up to 475	Up to 500	Up to 525	Up to 550	Up to 575	Up to 600
Plates																						
IS: 2002-1962	I	A	37	0.55R ₂₀	26	12.3	12.3	11.5	10.4	9.5	8.7	7.8	7.5	7.2	5.9	4.3	3.6	—	—	—	—	—
IS: 2002-1962	2A	A	42	0.50R ₂₀	25	14.0	12.9	11.9	10.8	9.8	9.0	8.1	7.7	7.4	5.9	4.3	3.6	—	—	—	—	—
IS: 2002-1962	2B	A	52	0.50R ₂₀	20	17.3	15.9	14.7	13.3	12.1	11.1	10.0	9.5	8.3	5.9	4.3	3.6	—	—	—	—	—
IS: 2041-1962	20Mo55	B	48	28	20	16.0	16.0	16.0	15.3	14.3	13.2	12.3	11.9	11.5	11.2	10.8	7.7	5.6	3.7	—	—	—
IS: 2041-1962	20Mn2	A	52	30	20	17.3	17.3	17.0	15.4	14.0	12.8	11.6	11.0	8.3	5.9	4.3	3.6	—	—	—	—	—
IS: 1570-1961	15Cr90Mo55	D	50	30	20	16.6	16.6	16.6	16.6	16.0	15.2	14.4	13.8	13.4	13.0	12.6	11.7	8.6	5.8	3.5	—	—
IS: 1570-1961	C15Mn75	A	42	23	25	14.0	14.0	13.0	11.8	10.7	9.8	8.9	8.4	8.1	5.9	4.3	3.6	—	—	—	—	—
Forgings																						
IS: 2004-1962	Class I	A	37	0.50R ₂₀	—	12.3	11.3	10.4	9.5	8.6	7.9	7.1	6.8	6.3	5.9	4.3	3.6	—	—	—	—	—
IS: 2004-1962	Class 2	A	44	0.50R ₂₀	15	14.6	13.4	12.4	11.3	10.2	9.3	8.5	8.0	7.7	5.9	4.3	3.6	—	—	—	—	—
IS: 2004-1962	Class 3	A	50	0.50R ₂₀	21	16.6	15.3	14.1	12.8	11.7	10.7	9.6	9.1	8.3	5.9	4.3	3.6	—	—	—	—	—
IS: 2004-1962	Class 4	A	63	0.50R ₂₀	15	21.0	19.3	17.8	16.1	14.7	13.4	12.2	11.5	8.3	5.9	4.3	3.6	—	—	—	—	—
IS: 1570-1961	20Mo55	B	48	28	20	16.0	16.0	16.0	15.3	14.3	13.2	12.3	11.9	11.5	11.2	10.8	7.7	5.6	3.7	—	—	—
IS: 2611-1964	15Cr90Mo55	D	50	30	20	16.6	16.6	16.6	16.6	16.0	15.2	14.4	13.8	13.4	13.0	12.6	11.7	8.6	5.8	3.5	—	—
IS: 1570-1961	10Cr2Mo1	E	50	32	20	16.6	16.6	16.6	16.6	16.6	16.6	16.4	16.1	15.8	15.3	14.9	12.7	9.6	7.0	4.9	3.2	2.3
Tubes, Pipes																						
IS: 3609-1966	1%Cr - 1%Mo Tube Normalized and Tempered	D	44	24	950/R ₂₀	14.6	14.6	14.2	13.6	12.8	12.1	11.5	11.1	10.7	10.4	10.0	9.7	8.6	5.8	3.5	—	—
IS: 3609-1966	2%Cr - 1%Mo Tube Normalized and Tempered	E	49	25	950/R ₂₀	16.3	15.6	15.0	14.5	14.0	13.5	12.8	12.6	12.3	12.0	11.6	11.3	9.6	7.0	4.9	—	—
IS: 1570-1961	20Mo55	B	46	25	950/R ₂₀	15.3	15.3	14.6	13.6	12.8	11.8	11.0	10.6	10.3	10.0	9.6	7.7	5.6	3.7	—	—	—
IS: 1914-1961	32 kgf/mm ² , Min Tensile Strength	A	32	0.50R ₂₀	950/R ₂₀	10.6	9.8	9.0	8.2	7.4	6.8	6.2	5.8	5.6	5.0	4.3	3.6	—	—	—	—	—
IS: 1914-1961	43 kgf/mm ² , Min Tensile Strength	A	43	0.50R ₂₀	950/R ₂₀	14.3	13.1	12.2	11.0	10.0	9.2	8.3	7.9	7.6	5.9	4.3	3.6	—	—	—	—	—
IS: 2416-1963	32 kgf/mm ² , Min Tensile Strength	A	32	0.50R ₂₀	950/R ₂₀	10.6	9.8	9.0	8.2	7.4	6.8	6.2	5.8	5.6	5.0	4.3	3.6	—	—	—	—	—

(Continued)

TABLE A.1 ALLOWABLE STRESS VALUES FOR CARBON AND LOW ALLOY STEEL IN TENSION — Contd

MATERIAL SPECIFICATION	GRADE OR DESIGNATION	CATEGORY OF CHEMICAL COMPOSITION (see TABLE B.1)	MECHANICAL PROPERTIES			ALLOWABLE STRESS VALUES IN kgf/mm ² AT DESIGN TEMPERATURE, °C																
			Tensile Strength <i>Min</i> kgf/mm ² R ₂₀	Yield Stress <i>Min</i> kgf/mm ² E ₂₀	Percentage Elongation <i>Min</i> on Gauge Length = 5.65√S ₀	Up to 50	Up to 100	Up to 150	Up to 200	Up to 250	Up to 300	Up to 350	Up to 375	Up to 400	Up to 425	Up to 450	Up to 475	Up to 500	Up to 525	Up to 550	Up to 575	Up to 600
IS : 1978-1961	St 18	A	31.6	17.6	—	10.5	10.5	9.9	9.0	8.2	7.5	6.7	6.4	6.2	5.9	4.3	3.6	—	—	—	—	—
	St 20	A	33.7	19.7	—	11.2	11.2	11.2	10.1	9.2	8.4	7.6	7.2	6.9	5.9	4.3	3.6	—	—	—	—	—
	St 21	A	33.7	21.1	—	11.2	11.2	11.2	10.8	9.8	9.0	8.1	7.7	7.4	5.9	4.3	3.6	—	—	—	—	—
	St 25	A	42.2	24.6	—	14.0	14.0	13.9	12.6	11.5	10.5	9.5	9.0	8.3	5.9	4.3	3.6	—	—	—	—	—
IS : 1979-1961	St 30	A	42.2	29.5	—	14.0	14.0	14.0	14.0	13.8	12.6	11.5	10.8	8.3	5.9	4.3	3.6	—	—	—	—	—
	St 32	A	44.3	32.3	—	14.7	14.7	14.7	14.7	13.8	12.5	11.8	8.3	5.9	4.3	3.6	—	—	—	—	—	—
	St 37	A	46.4	36.6	—	15.4	15.4	15.4	15.4	15.4	15.4	14.1	13.4	8.3	5.9	4.3	3.6	—	—	—	—	—
Castings*																						
IS : 3038-1965	Grade 1	A	55	35	17	13.7	13.7	13.7	13.5	12.2	11.2	10.1	9.6	6.2	4.4	3.2	2.7	—	—	—	—	—
	Grade 2	B	47	25	17	11.7	11.6	11.0	10.2	9.6	8.8	8.2	8.0	7.7	7.5	7.2	5.8	4.2	2.8	—	—	—
	Grade 3	C	52	31	15	13.0	13.0	13.0	12.7	11.9	11.0	10.2	9.9	9.6	9.3	8.4	5.8	4.2	2.8	—	—	—
	Grade 4	D	49	28	17	12.2	12.2	12.2	11.9	11.2	10.6	10.1	9.7	9.3	9.1	8.8	8.5	6.5	4.4	2.6	—	—
	Grade 5	E	52	31	17	13.0	13.0	13.0	13.0	13.0	12.5	11.9	11.7	11.4	11.1	10.8	9.5	7.2	5.3	3.7	2.4	—
	Grade 6	F	63	43	15	15.7	15.7	15.7	15.7	15.7	15.7	15.5	14.9	14.4	14.0	13.5	6.7	4.9	3.5	2.6	1.7	0.9
IS : 2856-1964	C Sw-C20	A	42	21	20	10.5	9.6	8.9	8.0	7.3	6.7	6.1	5.7	5.5	4.4	3.2	2.7	1.6	—	—	—	—
	C Sw-C25	A	49	25	18	12.2	11.5	10.6	9.6	8.7	8.0	7.2	6.8	6.2	4.4	3.2	2.7	1.6	—	—	—	—
Rivet and Stay Bars																						
IS : 1990-1962		A	37	0.55R ₂₀	26	12.3	12.3	11.5	10.4	8.6	7.9	7.1	6.8	6.5	5.9	4.3	3.6	—	—	—	—	—
		A	42	0.55R ₂₀	23	14.0	14.0	13.1	11.8	9.8	9.0	8.1	7.9	7.4	5.9	4.3	3.6	—	—	—	—	—
Sections, Plates, Bars																						
IS : 226-1962	St 42-S	A	42	24	23	14.0	14.0	13.6	12.3	9.8	9.0	8.1	—	—	—	—	—	—	—	—	—	—
IS : 961-1962	St 55 HTW	A	50	29	20	16.6	16.6	16.4	14.8	11.7	10.7	9.6	—	—	—	—	—	—	—	—	—	—
IS : 2062-1962	St 42-W	A	42	23	23	14.0	14.0	13.0	11.8	9.8	9.0	8.1	—	—	—	—	—	—	—	—	—	—
IS : 3039-1965	Grade A	A	—	—	—	14.0	14.0	13.0	11.8	9.8	9.0	8.1	—	—	—	—	—	—	—	—	—	—
	Grade D	A	—	—	—	14.6	16.6	16.4	14.8	11.7	10.7	9.6	—	—	—	—	—	—	—	—	—	—
IS : 3503-1966	Grade 1	A	37	0.50R ₂₀	26	12.3	11.3	10.4	9.5	8.6	7.9	7.1	6.8	6.5	5.9	4.3	3.6	—	—	—	—	—
	Grade 2	A	42	0.50R ₂₀	25	14.0	12.9	11.9	10.8	9.8	9.0	8.1	7.7	7.4	5.9	4.3	3.6	—	—	—	—	—
	Grade 3	A	44	0.50R ₂₀	23	14.6	13.4	12.4	11.3	10.2	9.3	8.5	8.0	7.7	5.9	4.3	3.6	—	—	—	—	—
	Grade 4	A	47	0.50R ₂₀	22	15.6	14.4	13.3	12.1	11.0	10.0	9.1	8.3	8.3	5.9	4.3	3.6	—	—	—	—	—
	Grade 5	A	50	0.50R ₂₀	21	16.6	15.3	14.1	12.8	11.7	10.7	9.6	9.1	8.3	5.9	4.3	3.6	—	—	—	—	—
IS : 3945-1966	Grade A-N	A	44	24	23	14.6	14.6	13.6	12.3	9.8	9.0	8.1	—	—	—	—	—	—	—	—	—	—
	Grade B-N	A	50	28.5	20	16.6	16.6	16.1	14.6	11.7	10.7	9.6	—	—	—	—	—	—	—	—	—	—

*These values have been based on a quality factor of 0.75. For additional inspection as detailed in Note to Table 2.1, a quality factor of 0.9 shall be used and the above stress values increased proportionally.